

UNIT – IV

OBJECTIVE:

To familiarize the students with the thermodynamics properties and its relations and uses of this thermodynamics relations

HELM HOLTZ AND GIBBS FUNCTIONS:

The combination of properties (U-TS) is also a Thermodynamic property. It is named as the Helmholtz function. It is denoted as A and a.

$$A = U - TS$$

$$a = u - Ts$$

Another combination of properties (H – Ts) is also a Thermodynamic property. It is named as Gibbs function. It is denoted as G and g.

$$G = H - TS$$

$$g = h - Ts$$

Mathematical Conditions for Exact Differentials (Properties)

Properties are mathematically exact differentials. These are point functions and their differences between two points do not depend upon the path of integration.

Let Z is function of two variables x and y

$$Z = Z(x,y)$$

$$dz = \left(\frac{\partial z}{\partial x}\right)_y dx + \left(\frac{\partial z}{\partial y}\right)_x dy \quad \text{-----} \quad (1)$$

$$\text{But } \left(\frac{\partial z}{\partial x}\right)_y = M \text{ and } \left(\frac{\partial z}{\partial y}\right)_x = N.$$

Then Eq. (1) Becomes

$$dz = Mdx + Ndy \quad \text{-----} \quad (2)$$

If x,y,z are all pint functions. Is there are state variable representing properties which do not depend on the path, then dx, dy and dz are exact differentials. Two mathematical conditions apply,

$$a) \left(\frac{\partial m}{\partial y} \right)_x = \left(\frac{\partial n}{\partial x} \right)_y$$

$$b) \left(\frac{\partial x}{\partial y} \right)_z = \left(\frac{\partial y}{\partial z} \right)_x \left(\frac{\partial z}{\partial x} \right)_y = -1$$

Among the variable x,y and z, any one variable may be considered as a function of the other two

$$x = x(y,z).$$

$$dx = \left(\frac{\partial x}{\partial y} \right)_z dy + \left(\frac{\partial x}{\partial z} \right)_y dz \text{ ----- (1)}$$

$$dz = \left(\frac{\partial z}{\partial x} \right)_y dx + \left(\frac{\partial z}{\partial y} \right)_x dy \text{----- (2)}$$

From (1) & (2)

$$\begin{aligned} dx &= \left(\frac{\partial x}{\partial y} \right)_z dy + \left(\frac{\partial x}{\partial z} \right)_y dz \left[\left(\frac{\partial z}{\partial x} \right)_y dx + \left(\frac{\partial z}{\partial y} \right)_x dy \right] \\ &= \left[\left(\frac{\partial x}{\partial y} \right)_z + \left(\frac{\partial x}{\partial z} \right)_y \left(\frac{\partial z}{\partial y} \right)_x \right] dy + \left(\frac{\partial x}{\partial z} \right)_y \left(\frac{\partial z}{\partial x} \right)_y dx \\ &= \left[\left(\frac{\partial x}{\partial y} \right)_z + \left(\frac{\partial x}{\partial z} \right)_y \left(\frac{\partial z}{\partial y} \right)_x \right] dy + dx \end{aligned}$$

$$\Rightarrow \left(\frac{\partial x}{\partial y} \right)_z + \left(\frac{\partial x}{\partial z} \right)_y \left(\frac{\partial z}{\partial y} \right)_x = 0$$

(OR)

$$\left(\frac{\partial x}{\partial y} \right)_z + \left(\frac{\partial z}{\partial x} \right)_y \left(\frac{\partial y}{\partial z} \right)_x = -1 \text{ ----- (3)}$$

Among the thermodynamic variable P, V and T, the following relation holds good

$$\left(\frac{\partial p}{\partial v}\right)_T + \left(\frac{\partial v}{\partial T}\right)_p \left(\frac{\partial T}{\partial p}\right)_v = -1 \quad (4)$$

Maxwell Relations:

Consider a simple compressible control mass of fixed chemical composition. The Maxwell relations, which can be written for such a system, are four equations relating the properties P,V,T and S.

The Maxwell relations are most easily derived by considering the different forms of the thermodynamic property relation.

$$du = Tds - pdv \quad (1)$$

and

$$dh = Tds + vdp \quad (2)$$

The above expressions are of the form

$U = u(s,v)$, $h=h(s,p)$ in both of which entropy is used as one of the two independent properties.

To eliminate entropy as an independent property by introducing two new properties and thereby two new forms of the thermodynamic property relations.

The first of these is the helmholtz function A,

$$A = U - Ts, \quad a = u - Ts \quad (3)$$

Differentiate eq (3) we get

$$da = du - Tds - sdT \quad (4)$$

Put eqn (1) in (4) we get

$$\begin{aligned}\Rightarrow da &= Td/s - pdv - Td/s - sdT \\ &= -sdT - pdv\end{aligned}\quad \text{-----} \quad (5)$$

The above expression is a form of the property relation utilizing T and v as the independent properties.

The second new property is the Gibbs function G.

$$G = H - TS \quad , \quad g = h - Ts \quad \text{-----} \quad (6)$$

Differentiate eq (2) we get

$$dg = dh - Tds - sdT \quad \text{-----} \quad (7)$$

put eq (7) in eq (2) we get

$$\begin{aligned}\Rightarrow dg &= Tds + vdp - Tds - sdT \\ &= -sdT + vdp\end{aligned}\quad \text{-----} \quad (8)$$

Eq. (1), (2), (5), (8) all relations involving only properties. We conclude that these are exact differentials.

$$dz = m dx + n dy$$

The coefficient M and n are

$$\left(\frac{\partial m}{\partial y}\right)_x = \left(\frac{\partial n}{\partial x}\right)_y$$

From

Eq.(1) we get

$$\left(\frac{\partial T}{\partial u}\right)_s = -\left(\frac{\partial p}{\partial s}\right)_v \quad \text{-----} \quad (9)$$

Eq.(2) we get

$$\left(\frac{\partial T}{\partial u}\right)_s = \left(\frac{\partial v}{\partial s}\right)_p \quad \text{-----} \quad (10)$$

Eq.(5) we get

$$\left(\frac{\partial p}{\partial T}\right)_v = \left(\frac{\partial v}{\partial s}\right)_T \text{-----} \quad (11)$$

Eq.(8) we get

$$\left(\frac{\partial v}{\partial T}\right)_p = \left(\frac{\partial s}{\partial p}\right)_T \text{-----} \quad (12)$$

These four equations are known as Maxwell relations for a simple compressible mass.

Clausius Clapeyron Equation:

An important application of one of the Maxwell relations is in the derivation of the clapeyron equation. This equation relates the enthalpy of vaporization of a substance with the slope of its vapour pressure curve.

During phase transitions like melting, vaporization and sublimation the temperature and pressure remain constant while the entropy and volume changes. If x is the fraction of initial phase i which has been transformed into final phase f .

$$\left. \begin{aligned} S &= (1-x)s(i) + s(f) \\ V &= (1-x)v(i) + v(f) \end{aligned} \right\} \text{-----} \quad (9)$$

where s and v are

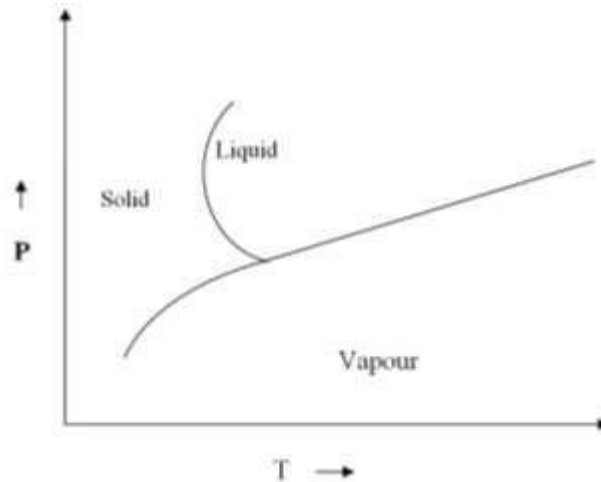
Linear functions of x .

This is used to find the value of the property which is not measurable for pure substances. This equation provides a basis for calculation of properties in a two phase region.

Consider a change of phase of pure substances from saturated liquid to saturated vapour. During the vaporization process both temperature and pressure remains constant, but specific volume and entropy changes.

$$S = f(T, V)$$

$$\Rightarrow ds = \left(\frac{\partial s}{\partial T} \right)_v dT + \left(\frac{\partial s}{\partial v} \right)_T dv \quad \text{-----} \quad (1)$$



P-T diagram for a pure substance

When the phase is changing from saturated liquid to Sat.vapour, temperature relations const.

Eq. (1) becomes.

$$ds = \left(\frac{\partial s}{\partial v} \right)_v dT \quad \text{-----} \quad (2)$$

From Maxwell equation gives.

$$\left(\frac{\partial s}{\partial v} \right)_T = \left(\frac{\partial p}{\partial T} \right)_v \quad \text{-----} \quad (3)$$

Put (3) in (2) we get

$$\Rightarrow ds = \left(\frac{\partial p}{\partial T} \right)_v dv \quad \text{-----} \quad (4)$$

$\left(\frac{\partial p}{\partial T} \right)_v$ is the slope of the saturation curve.

During a phase change process, the pressure is a sat. Pressure which depends on temperature (T) only and independent of specific volume (v)

Integrating the eq.(4) we get

$$\int_{sf}^{sg} ds = \frac{\partial p}{\partial T} \int_{vf}^{vg} dv$$

$$sg - sf = \frac{dp}{dT} (vg - vf)$$

$$\Rightarrow \frac{dp}{dT} = \frac{sg - sf}{vg - vf}$$

$$\Rightarrow \frac{dp}{dT} = \frac{sfg}{vgf} \text{-----} \quad (5)$$

From second law of thermodynamics.

$$ds = \frac{dQ}{T} \quad \text{but } dQ = dH \text{ for const. press. process.}$$

$$\Rightarrow ds = \frac{dH}{T} \text{-----} \quad (6)$$

Integrating we get

$$\int_{sf}^{sg} ds = \int_{hf}^{hg} \frac{dH}{T}$$

$$sg - sf = \frac{1}{T} [hg - hf]$$

$$sfg = \frac{hfg}{T} \text{-----} \quad (7)$$

put eq (7) in (5)

$$\left(\frac{\partial p}{\partial T} \right)_{\text{sat}} = \frac{hfg}{T vfg} \text{-----} \quad (8)$$

The above equation is known as clapeyron equation. This equation can be used to determine the enthalpy of evaporation at a given temp if the slope of the saturation curve on a p-T diagram for the given fluid is known.

For very low pressure, v_f is very small

$$v_f \ll v_g$$

$$\text{Eq.(8)} \Rightarrow \left(\frac{dp}{dT} \right)_{\text{sat}} = \frac{h_{fg}}{T v_{fg}}$$

$$\Rightarrow \left(\frac{dp}{dT} \right)_{\text{sat}} = \frac{h_g - h_f}{T(v_g - v_f)}$$

$$v_f = 0$$

$$\Rightarrow \left(\frac{dp}{dT} \right)_{\text{sat}} = \frac{h_g - h_f}{T v_g} \quad \text{-----} \quad (9)$$

For ideal gas $p v = R T$

$$v = \frac{R T}{p}$$

$$\Rightarrow v_g = \frac{R T}{p} \quad \text{-----} \quad (10)$$

Put (10) in (9)

$$(9) \Rightarrow \left(\frac{dp}{dT} \right)_{\text{sat}} = \left(\frac{h_g - h_f}{\frac{R T}{p}} \right) \frac{1}{T}$$

$$\frac{dp}{dT} = \frac{p h_{fg}}{R T^2}$$

$$\frac{dp}{dT} = \frac{d T h_{fg}}{R T^2}$$

$$\int_{P_1}^{P_2} \frac{dP}{P} = \int_{T_1}^{T_2} \frac{dP}{R T^2} h_{fg}$$

$$\int_{P_1}^{P_2} \frac{dP}{P} = \frac{h_{fg}}{R} \int_{T_1}^{T_2} \frac{dT}{T^2}$$

$$\ln \left(\frac{dp}{dT} \right) = \frac{h_{fg}}{R} \left[\frac{-1}{T} \right]_{T_1}^{T_2}$$

$$\ln \left(\frac{dp}{dT} \right) = \frac{h_{fg}}{R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right] \quad \text{-----} \quad (11)$$

The above equation is known as Clausius - Clapeyron equation. It is also used to determine the variation in saturation pressure with temperature.

T – D Relations:

(i) Let entropy s be imagined as a function of T and V then.

$$ds = \left(\frac{\partial s}{\partial T} \right)_v dT + \left(\frac{\partial s}{\partial v} \right)_T dv$$

$$\Rightarrow Tds = T \left(\frac{\partial s}{\partial T} \right)_v dT + T \left(\frac{\partial s}{\partial v} \right)_T dv \quad \text{-----} \quad (1)$$

$$\Rightarrow Tds = C_v dT + T \left(\frac{\partial s}{\partial v} \right)_T dv \quad \text{-----} \quad (2)$$

And $\left(\frac{\partial s}{\partial v} \right)_T = \left(\frac{\partial p}{\partial T} \right)_v \quad \text{-----} \quad (3)$

Maxwell's third equation

put eq (2), (3) in eq (1)

$$\Rightarrow Tds = C_v dT + T \left(\frac{\partial p}{\partial T} \right)_v dV \quad \text{-----} \quad (4)$$

This equation is known as first Tds equation.

(ii) If entropy S is considered as a function of T and P .

Then $ds = \left(\frac{\partial s}{\partial T} \right)_p dT + \left(\frac{\partial s}{\partial p} \right)_T dp$

$$Tds = T \left(\frac{\partial s}{\partial T} \right)_p dT + T \left(\frac{\partial s}{\partial p} \right)_T dp \quad \text{-----} \quad (5)$$

$$\Rightarrow T \left(\frac{\partial s}{\partial T} \right)_p = C_p \quad \text{-----} \quad (6)$$

C_p – Specific heat at const. pressure.

And $\left(\frac{\partial s}{\partial p} \right)_T = - \left(\frac{\partial v}{\partial T} \right)_p \quad \text{-----} \quad (7)$

Maxwell's fourth equation.

Put (6) & (7) in eq (5) we get

$$Tds = c_p dT - T \left(\frac{\partial v}{\partial T} \right)_p dp \quad \text{-----} \quad (8)$$

Specific Heat Relations:

$$Tds = C_v dT + T \left(\frac{\partial p}{\partial T} \right)_v dv \quad \text{-----} \quad (1)$$

$$Tds = c_p dT - T \left(\frac{\partial v}{\partial T} \right)_p dp \quad \text{-----} \quad (2)$$

Equation eq (1) & (2) we get

$$C_p - C_v = - T \left(\frac{\partial v}{\partial T} \right)_p \left(\frac{\partial p}{\partial v} \right)_T \quad \text{-----} \quad (3)$$

This relation can be expressed in terms of two other thermodynamic properties called the volume expansivities β and the isothermal compressibility α .

$$\left. \begin{aligned} \beta &= \frac{1}{v} \left(\frac{\partial v}{\partial T} \right)_p \\ \alpha &= -\frac{1}{v} \left(\frac{\partial v}{\partial p} \right)_T \end{aligned} \right\} \quad \text{-----} \quad (4)$$

Put (4) in (3) we get

$$C_p - C_v = v T \beta^2 / \alpha \quad \text{-----} \quad (5)$$

- The isothermal compressibility α is a positive quantity for all substances in all phases.
- The volume expansivity could be negative for some substances, but in square is always positive
- The constant - pressure specific heat is always greater than or equal to the constant volume specific heat.
- $C_p \geq C_v$

- The difference between C_p and C_v approached Zero as the absolute temperature approaches zero.

GIBBS PHASE RULE:

$$f = c - \phi + 2$$

the above equation is known as the Gibbs phase rule for a non reactive system

where,

f = degree of freedom

c = chemical constituents

ϕ = phase.